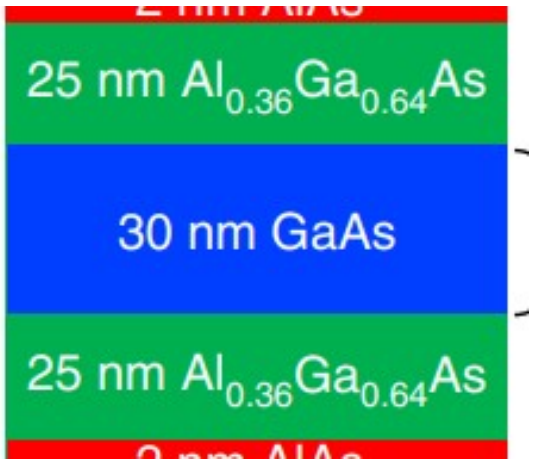
Electrons in a magnetic field, quantum Hall effect and Landau Levels

Materials showing quantum Hall effect

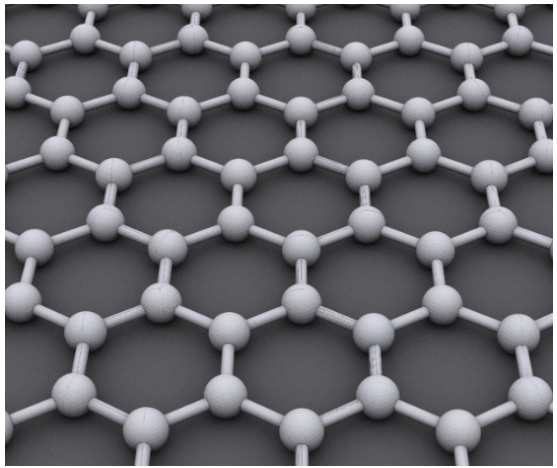
GaAs quantum wells



$$E \sim B$$

$$T \sim 1K$$

Graphene

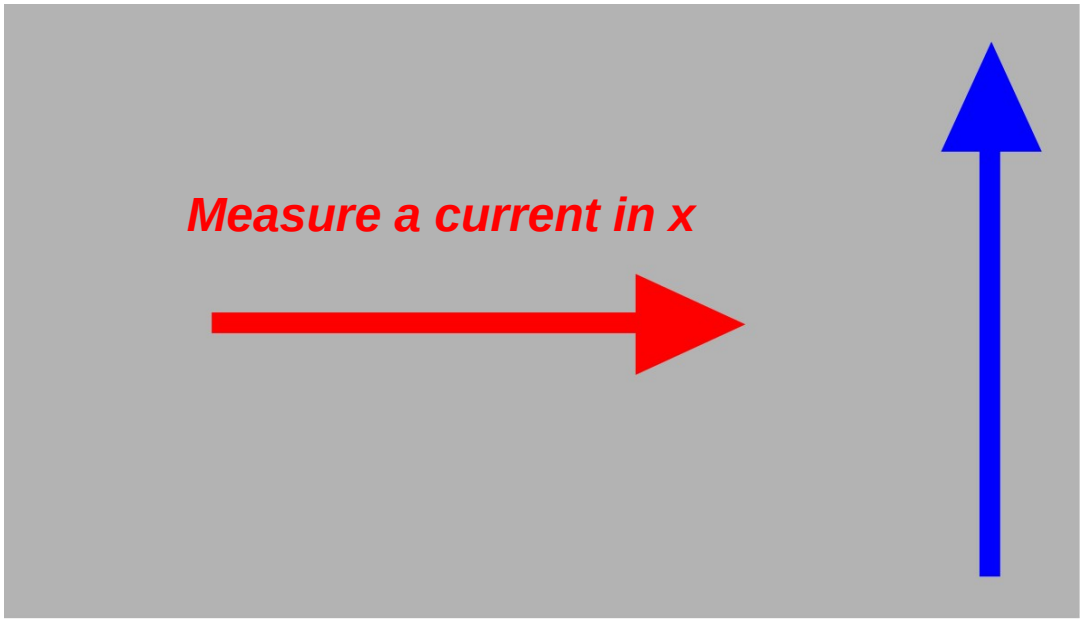


$$E \sim \sqrt{B}$$

$$T \sim 100K$$

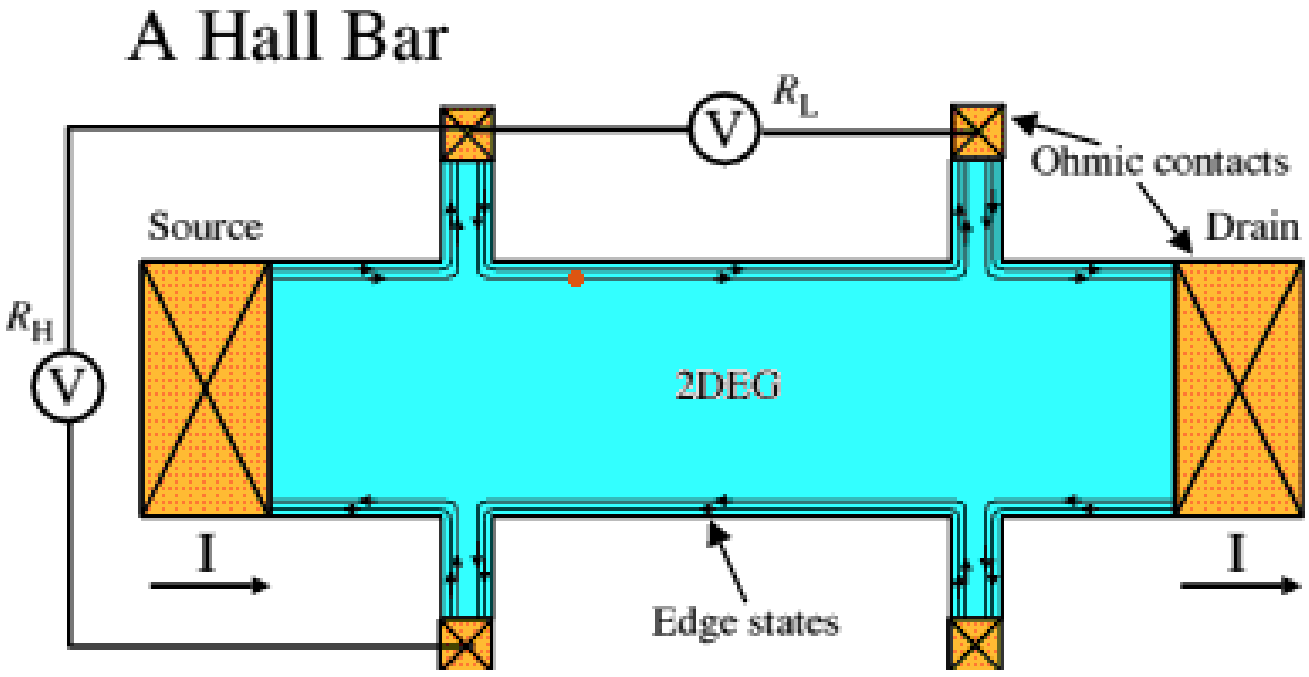
The quantum Hall state

Take a two-dimensional material

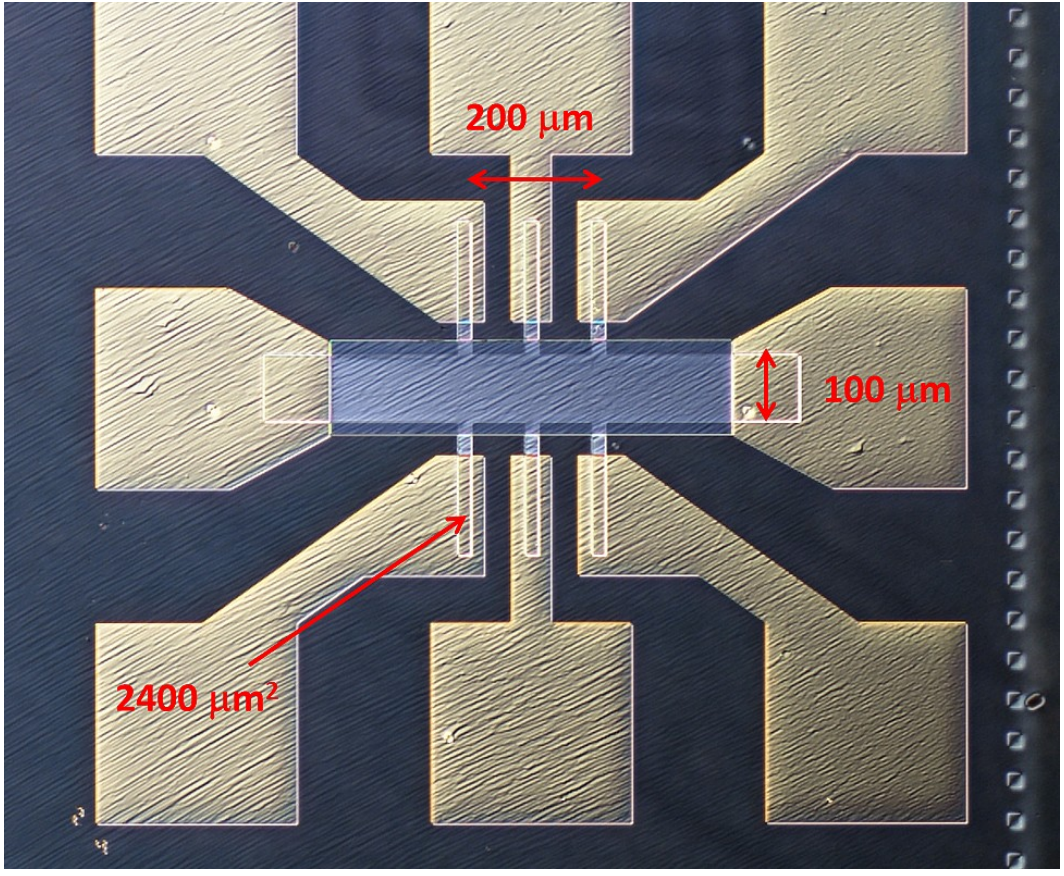


Hall conductance $J_x = \sigma_{xy} V_y$

Quantum Hall devices



Quantum Hall devices



Reminder: Linear response for transverse current

$$J_x = \sigma_{xy} V_y$$

The Hall conductivity is obtained as $\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k}$

with $\Omega_\alpha(\mathbf{k}) = i \sum_{\beta \neq \alpha} \frac{\langle \Psi_\alpha | \partial H/\partial k_x | \Psi_\beta \rangle \langle \Psi_\beta | \partial H/\partial k_y | \Psi_\alpha \rangle}{(\epsilon_\alpha - \epsilon_\beta)^2} - \alpha \leftrightarrow \beta$

Berry curvature of a band

Expression coming from perturbation theory

The Hall conductivity

The Hall conductivity is obtained as $\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k}$

Using $\langle \Psi_\alpha | \partial H / \partial k_\mu | \Psi_\beta \rangle = \langle \partial_{k_\mu} \Psi_\alpha | \Psi_\beta \rangle (\epsilon_\alpha - \epsilon_\beta)$

the Hall conductivity can be expressed in terms of

Berry curvature

$$\Omega_\alpha = \partial_{k_x} A_y^\alpha - \partial_{k_y} A_x^\alpha$$

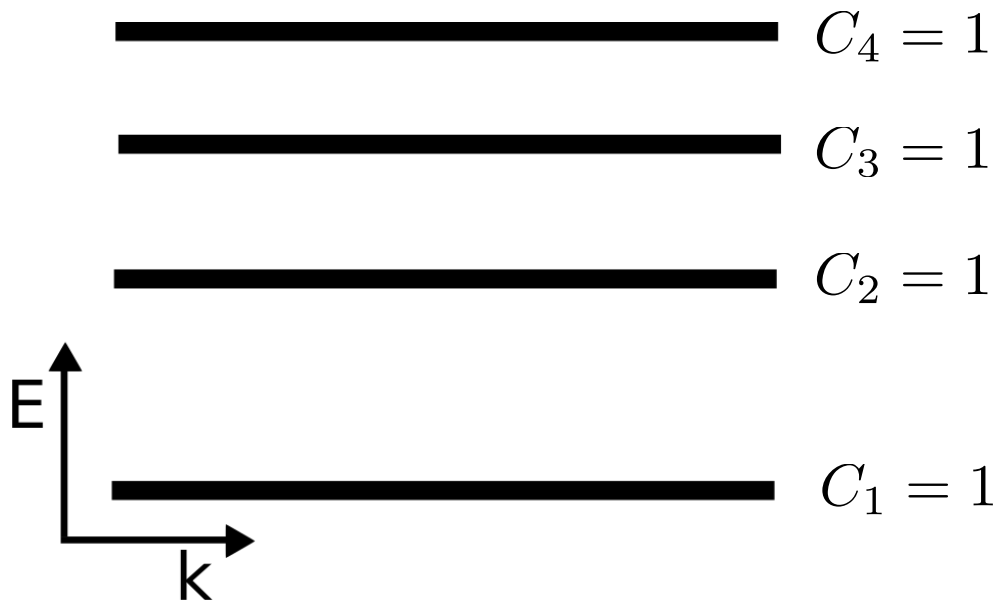
Berry connection

$$A_\mu^\alpha = i \langle \partial_{k_\mu} \Psi_\alpha | \Psi_\alpha \rangle$$

$$\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k} = \sum_\alpha C_\alpha = C$$ ← Chern number

Chern numbers in the quantum Hall state

Band-structure in the quantum Hall state



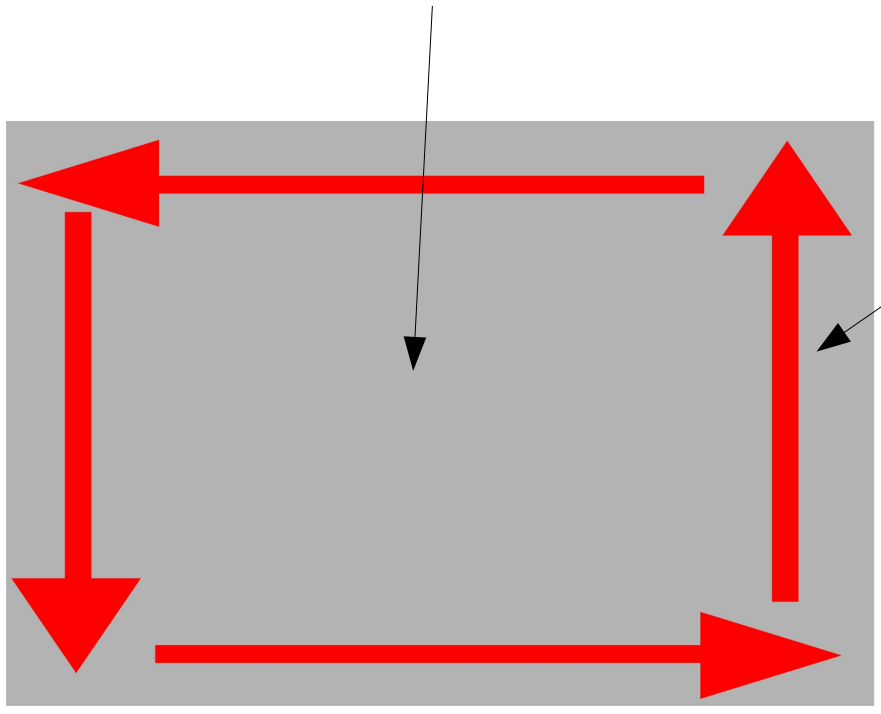
Hall conductivity

$$\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k} = \sum_\alpha C_\alpha = C$$

Each band (a.k.a Landau level), contributes with Chern number +1

The puzzling quantum Hall effect

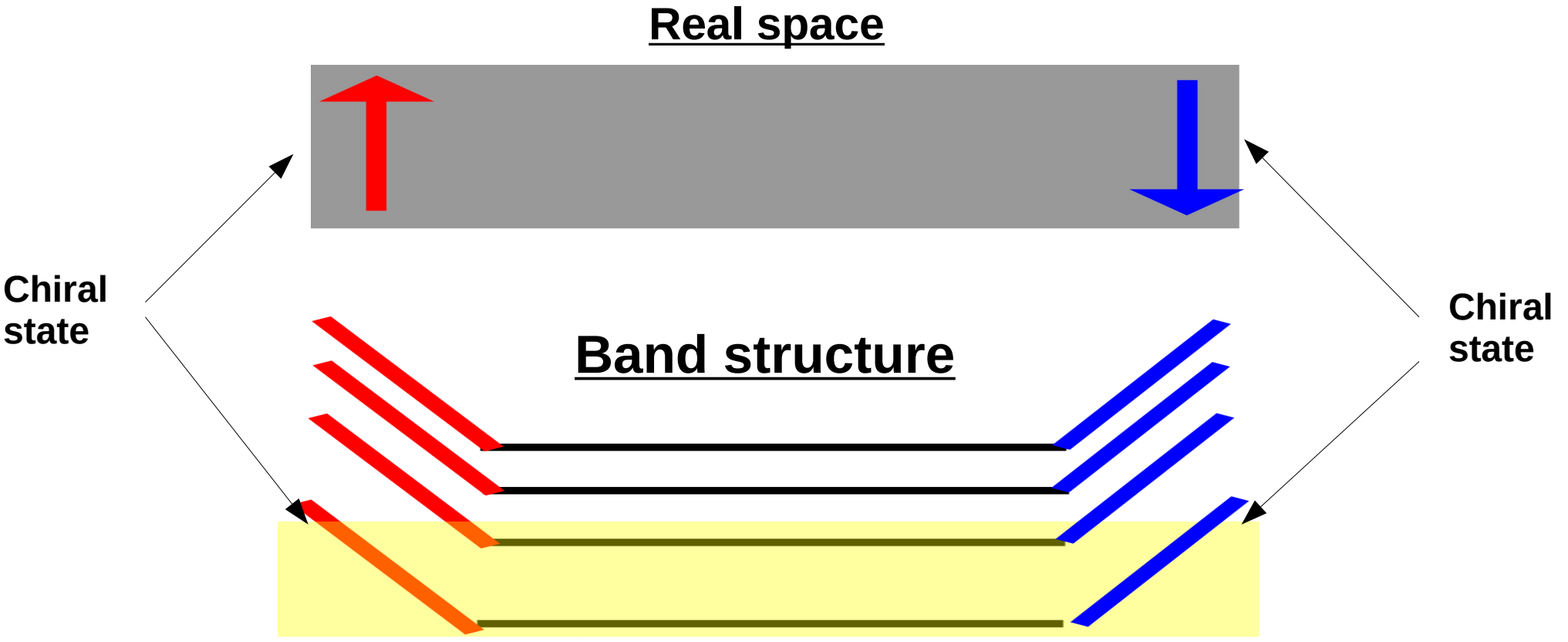
The bulk of a quantum Hall state is insulating



The edge has chiral states

Chiral: propagating only in one direction

The quantum Hall effect



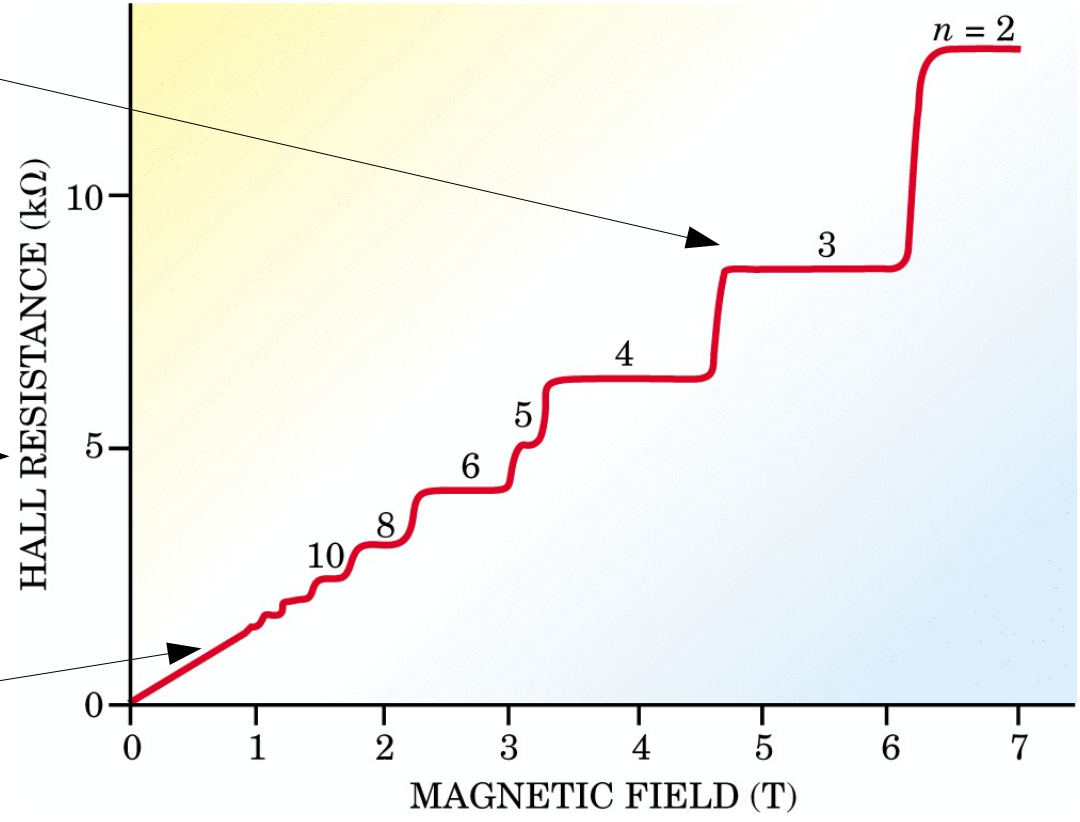
The quantum Hall state

Quantum regime

In the quantum Hall regime

$$\rho_{xy} \sim \sigma_{xy}^{-1}$$

Classical regime



Electrons in a magnetic field

Coupling (classical) electrons to a gauge field

The continuum limit

$$H = \frac{p^2}{2}$$

Given the kinetic energy without magnetic field

By replacing momentum by the canonical momentum
we recover the equations of motion in a magnetic field

$$H = \frac{\Pi^2}{2} \qquad \Pi = \mathbf{p} + \mathbf{A}$$

Canonical momentum

$$\mathbf{B} = \nabla \times \mathbf{A}$$

Magnetic field

Magnetic vector potential

Coupling (quantum) electrons to a gauge field

$$H = \frac{\Pi^2}{2} \qquad \Pi = \mathbf{p} + \mathbf{A}$$ Canonical momentum

$$p_\alpha = -i\partial_\alpha$$ Quantized momentum

Hamiltonian for electrons in a magnetic field

$$H = \frac{(\sum_\alpha -i\partial_\alpha + A_\alpha)^2}{2} \qquad \Psi(\mathbf{r})$$ Gauge #1

$$H = \frac{(\sum_\alpha -i\partial_\alpha)^2}{2} \qquad e^{i\int_0^{\mathbf{r}} \mathbf{A}(\mathbf{r}')d\mathbf{r}'}\Psi(\mathbf{r})$$ Gauge #2

Coupling electrons to a magnetic field in a tight binding model

Given a tight-binding model without magnetic field

$$H = \sum t_{ij} c_i^\dagger c_j$$

How do we rewrite it in the presence of a magnetic field?

$$\Psi(\mathbf{r}) \rightarrow e^{i \int_0^{\mathbf{r}} \mathbf{A}(\mathbf{r}') d\mathbf{r}'} \Psi(\mathbf{r})$$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

Coupling electrons to a magnetic field in a tight binding model

$$H = \sum t_{ij} c_i^\dagger c_j$$

$$\Psi(\mathbf{r}) \to e^{i \int_0^{\mathbf{r}} \mathbf{A}(\mathbf{r}') d\mathbf{r}'} \Psi(\mathbf{r})$$

Recall that, by definition

$$|\mathbf{r}_i\rangle = c_i^\dagger |\Omega\rangle \qquad t_{ij} = \langle \mathbf{r}_i | H | \mathbf{r}_j \rangle \qquad c_i |\Omega\rangle = 0$$

Coupling a gauge field becomes

$$|\mathbf{r}_i\rangle \to e^{i \int_0^{\mathbf{r}_i} \mathbf{A}(\mathbf{r}') d\mathbf{r}'} |\mathbf{r}_i\rangle$$

Which transforms the hoppings as

$$t_{ij} \to e^{i \int_{\mathbf{r}_i}^{\mathbf{r}_j} \mathbf{A}(\mathbf{r}') d\mathbf{r}'} t_{ij}$$

Two ways of coupling electrons to a magnetic field

Continuum limit

Schrodinger electrons $H = \frac{\Pi^2}{2}$

Dirac electrons $H = \sum_\alpha \Pi_\alpha \sigma_\alpha$

$$\Pi = \mathbf{p} + \mathbf{A}$$

Canonical momentum

Tight binding model

$$H = \sum t_{ij} c_i^\dagger c_j$$

$$t_{ij} \to e^{i\phi_{ij}} t_{ij}$$

$$\phi_{ij} = \int_{\mathbf{r}_i}^{\mathbf{r}_j} \mathbf{A}(\mathbf{r}') d\mathbf{r}'$$

Peierls substitution

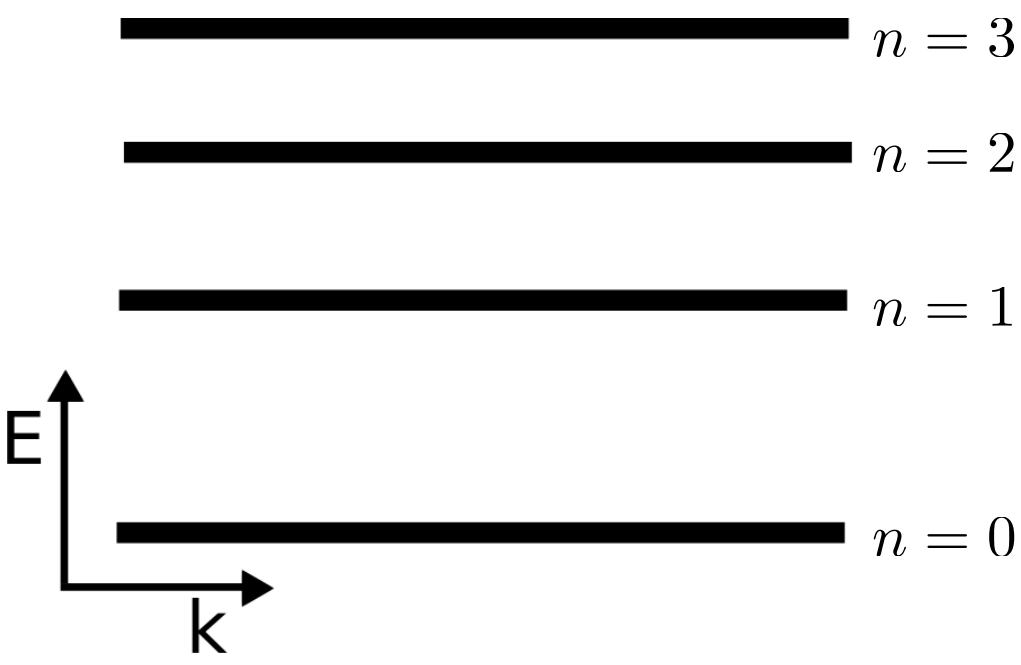
For typical materials

$\phi_{ij} \sim 10^{-4} \sim 10T$

Landau levels

Landau levels in a nutshell

Band-structure in the quantum Hall state



The energy levels are

$$E \sim \left(n + \frac{1}{2}\right) B$$

$$E \sim 1meV$$

For a Dirac equation they would be

$$E \sim \sqrt{nB}$$

$$E \sim 100meV$$

The gauge of the magnetic potential

Let us consider the simplest magnetic field

$$\mathbf{B} = (0, 0, B_z)$$

We can write down two different gauges for the magnetic potential $\mathbf{B} = \nabla \times \mathbf{A}$

Landau gauge

$$\mathbf{A} = (-B_z y, 0, 0)$$

Respects one translational symmetry

$$[p_x, A_\alpha] = 0$$

Symmetric gauge

$$\mathbf{A} = \frac{1}{2}(-B_z y, B_z x, 0)$$

Convenient for the fractional quantum Hall wavefunction

Electrons coupled to a magnetic field

Let us take a conventional electron gas coupled to a gauge field

$$H = \frac{(\sum_\alpha -i\partial_\alpha + A_\alpha)^2}{2}$$

Momentum

Gauge potential

Minimal gauge coupling

Landau levels of Schrodinger electrons in a nutshell

Lets take a quadratic Hamiltonian $H \sim p_x^2 + p_y^2$

And add a magnetic field (minimal coupling) $\mathbf{p} \to \mathbf{p} + \mathbf{A}$

Take the Landau gauge

$$\mathbf{A} = (0, -Bx, 0)$$
$$\nabla \times \mathbf{A} = (0, 0, B)$$

Plugging the magnetic potential in we get $H \sim p_x^2 + B^2 x^2$

(this looks like an harmonic oscillator)

Quantized levels in a magnetic field $E_n \sim nB$

Landau levels of Dirac electrons in a nutshell

Lets take a Dirac Hamiltonian

$$H \sim \begin{pmatrix} 0 & p_x + ip_y \\ p_x - ip_y & 0 \end{pmatrix}$$

And add a magnetic field (minimal coupling)

$$\mathbf{p} \to \mathbf{p} + \mathbf{A}$$

Take the Landau gauge

$$\mathbf{A} = (0, -Bx, 0)$$
$$\nabla \times \mathbf{A} = (0, 0, B)$$

Plugging the magnetic potential in we get

$$H^2 \sim p_x^2 + B^2 x^2$$

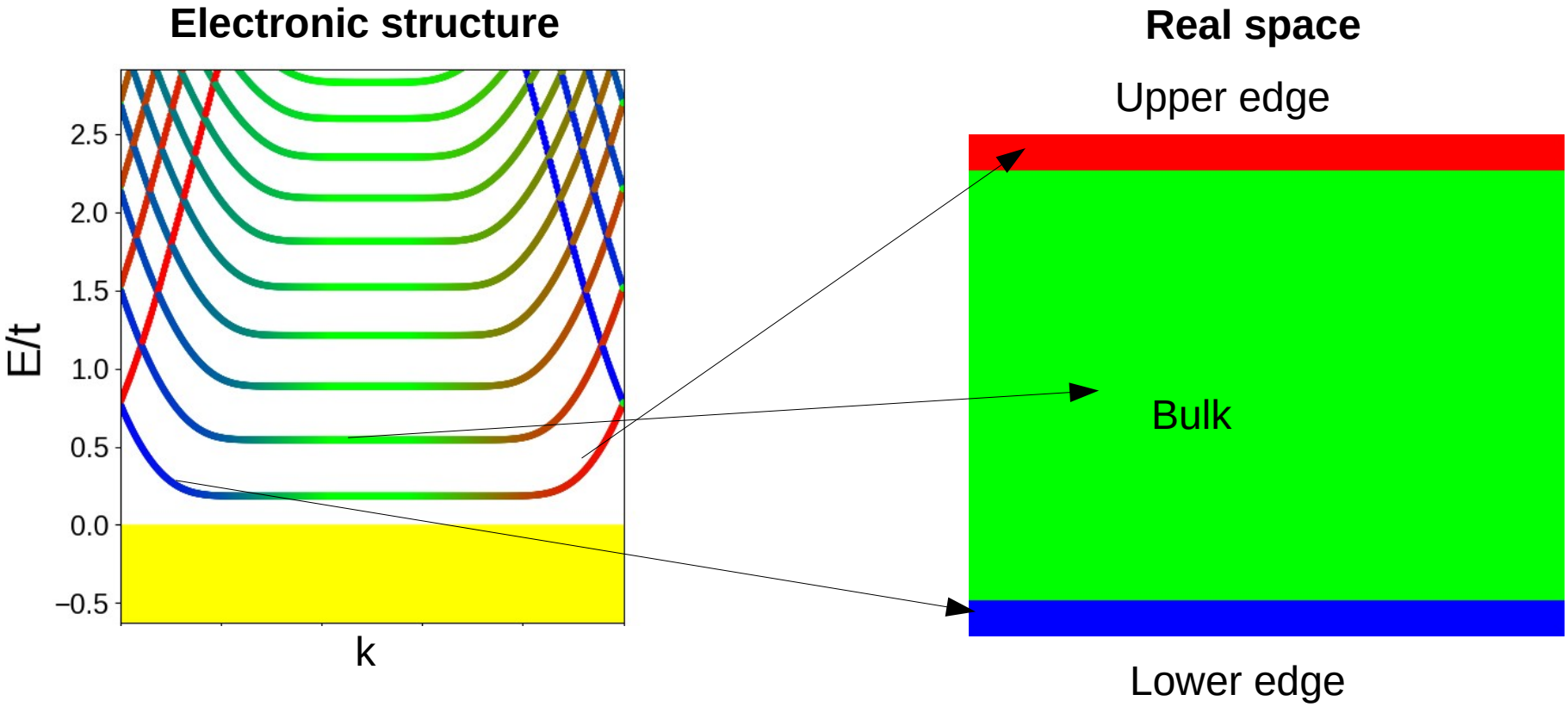
(this looks like an harmonic oscillator)

$$E_n^2 \sim nB$$

Quantized levels in a magnetic field

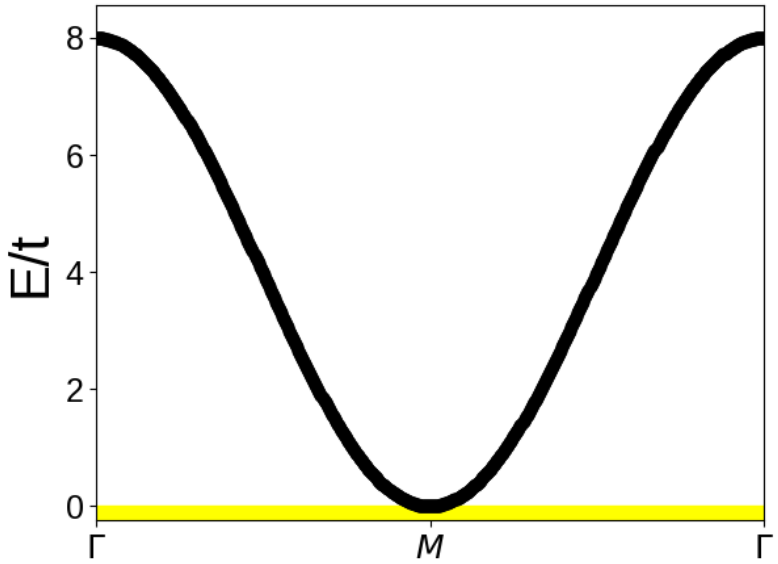
$$E_n \sim \sqrt{nB}$$

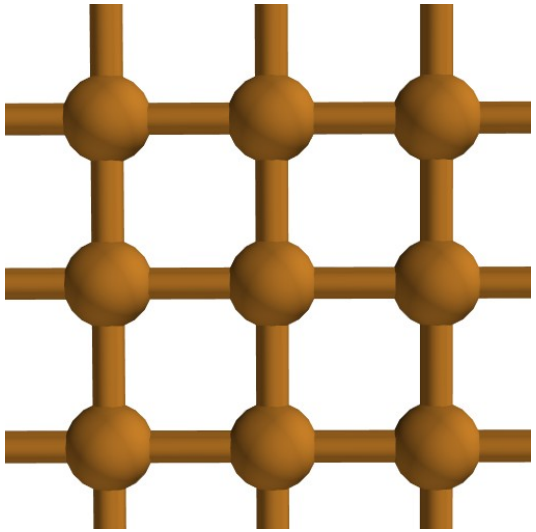
Landau levels in a nutshell



The Landau levels of quadratic electronic gas

Band structure of a square lattice

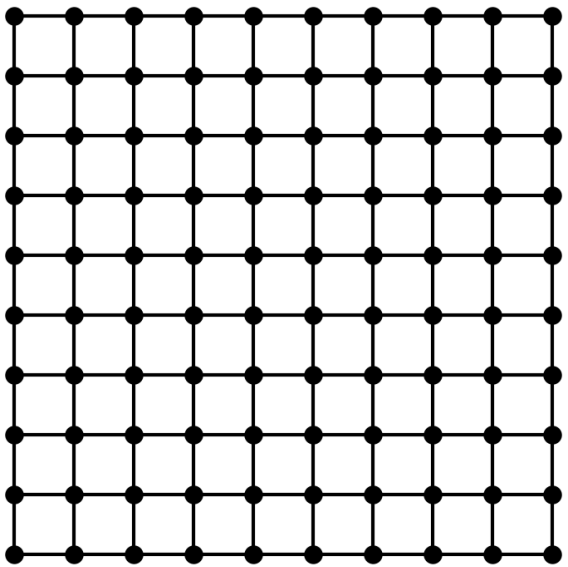




$$E \sim k_x^2 + k_y^2$$

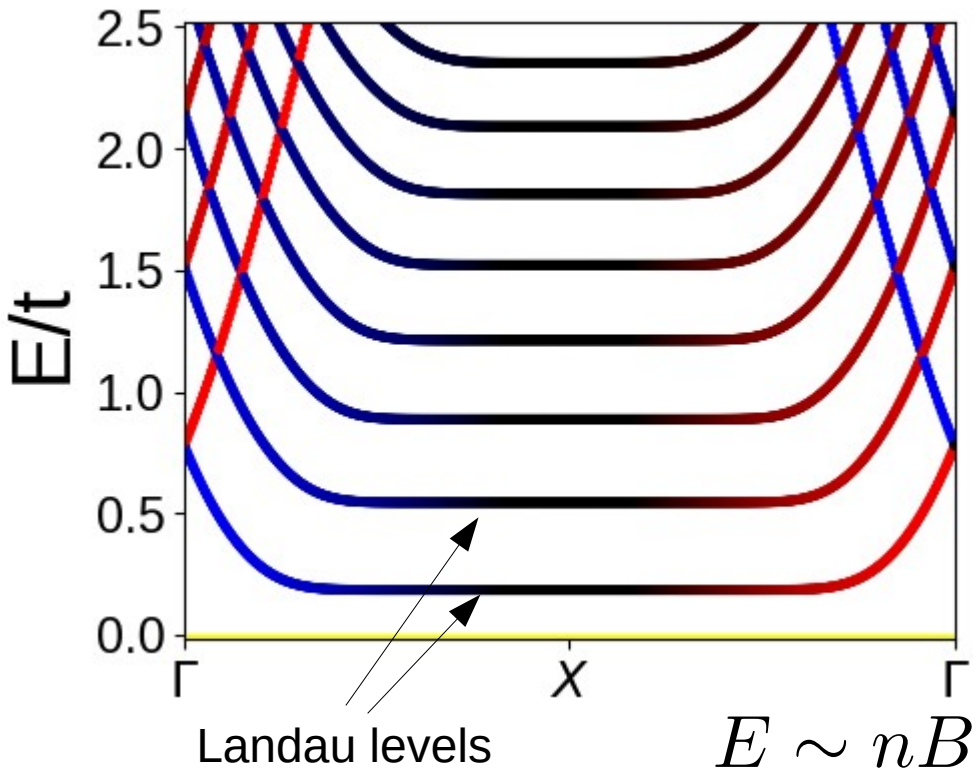
The Landau levels of quadratic electronic gas

Structure of a ribbon



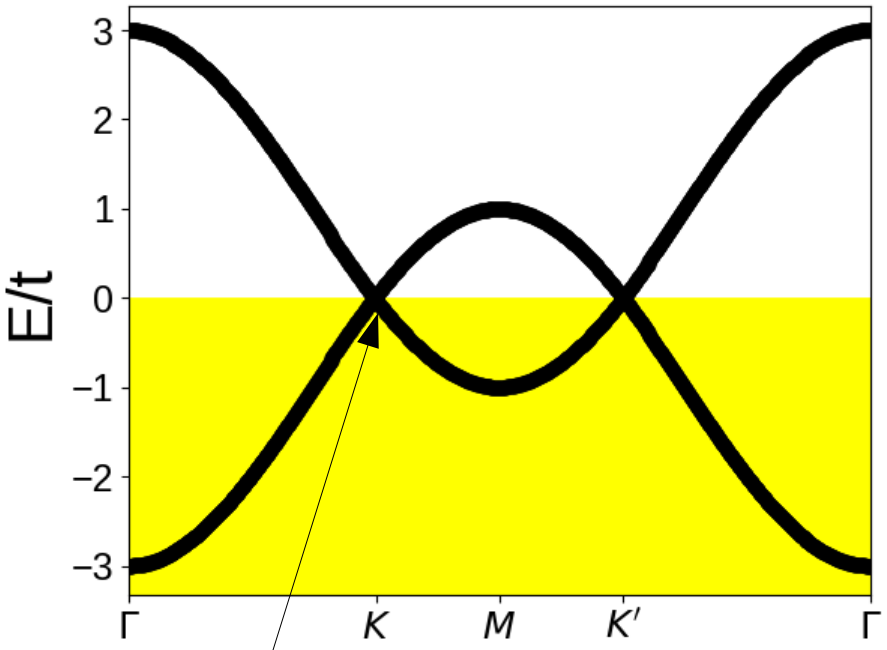
```
from pyqula import geometry,ribbon
g = geometry.square_lattice() # create the lattice
g = ribbon.bulk2ribbon(g,n=30) # create a ribbon
h = g.get_hamiltonian() # get the tight binding Hamiltonian
h.add_orbital_magnetic_field(0.03) # add magnetic field
h.add_onsite(4.) # shift Fermi energy
(k,e,c) = h.get_bands(operator="yposition") # get the band structure
```

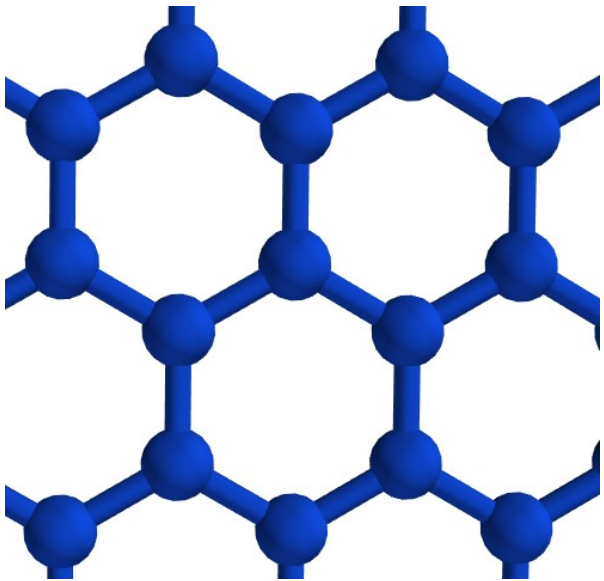
Band structure



The Landau levels of a Dirac electronic gas

Band structure of a honeycomb lattice

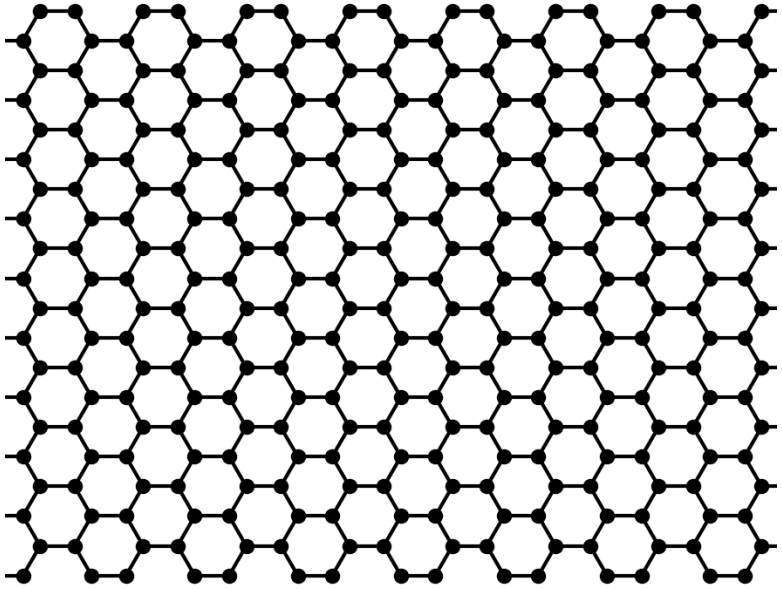




Effective Dirac equations

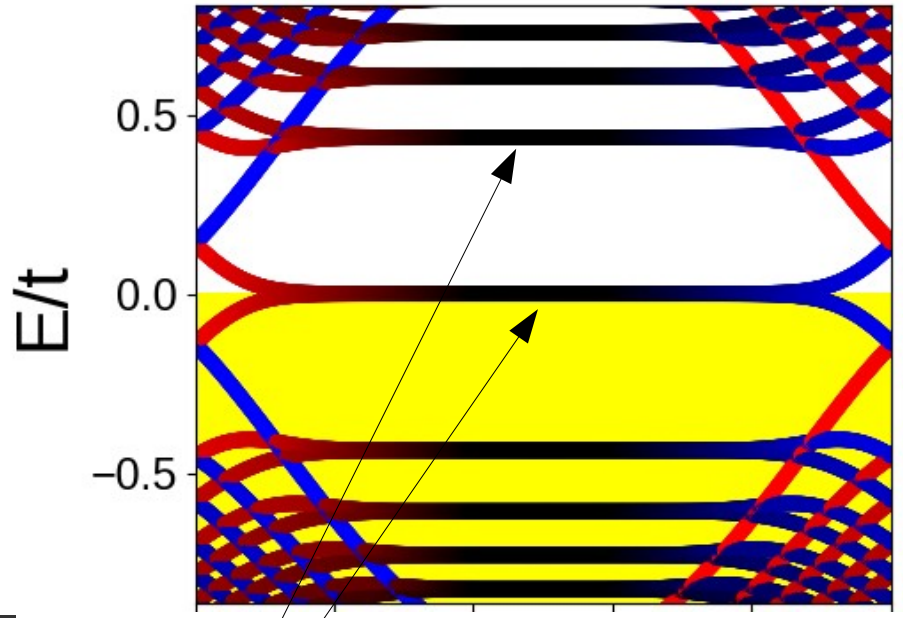
The Landau levels of a Dirac electronic gas

Structure of a ribbon



```
from pyqula import geometry
g = geometry.honeycomb_armchair_ribbon(30) # create the lattice
h = g.get_hamiltonian() # get the tight binding Hamiltonian
h.add_orbital_magnetic_field(0.007) # add magnetic field
(k,e,c) = h.get_bands(operator="yposition",kpath=["M","G","M"]) # get bands
```

Band structure



Landau levels

$$E \sim \sqrt{nB}$$

The many-body Landau level ground state

Ladder operators for Landau levels

Let us define new operators for the full Hamiltonian

$$a^\dagger = \frac{1}{\sqrt{2B}}\left((p_x - A_x) + i(p_y - A_y)\right)$$ "creation"

$$a = \frac{1}{\sqrt{2B}}\left((p_x - A_x) - i(p_y - A_y)\right)$$ "annihilation"

Canonical commutation relation

$$\nabla \times \mathbf{A} = (0, 0, B) \qquad aa^\dagger - a^\dagger a = 1$$

Ladder operators for Landau levels

The Hamiltonian for the bulk quantum Hall state can be written as

$$H = \frac{(\sum_\alpha -i\partial_\alpha + A_\alpha)^2}{2}$$

$$H = B\left(a^\dagger a + \frac{1}{2}\right)$$

$$\nabla \times \mathbf{A} = (0, 0, B)$$

$$aa^\dagger - a^\dagger a = 1$$

With eigenenergies

$$E_n = B\left(n + \frac{1}{2}\right)$$

The Landau levels in the symmetric gauge

Take the symmetric gauge

$$\mathbf{A} = \frac{B}{2}(-y, x, 0) \qquad \nabla \times \mathbf{A} = (0, 0, B)$$

0th Landau level wavefunctions $\Psi_m(z) \sim z^m e^{-|z|^2/(4\ell^2)}$

m is the angular momentum $z = x + iy \qquad \ell \sim 1/\sqrt{B}$

This wavefunction will be our starting point for the fractional quantum Hall state

Single-particle wavefunction

Landau level wavefunction

$$\Psi_m(z) \sim z^m e^{-|z|^2/(4\ell^2)}$$

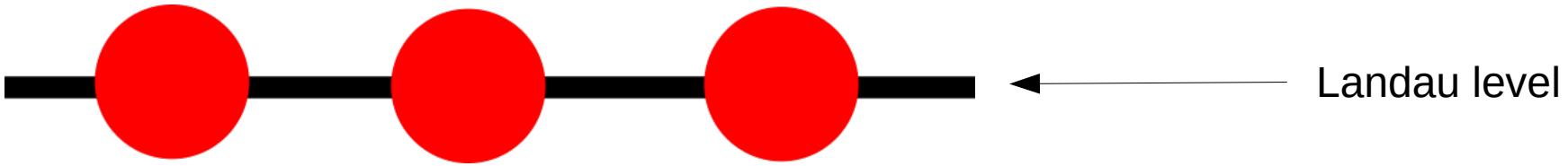
(another) Landau level wavefunction

$$\Psi(z) \sim p(z) e^{-|z|^2/(4\ell^2)}$$

Polynomial

This is for a single electron, how do we extend it to many-electrons?

The filled lowest Landau level



How to build the many-body wavefunction:
- Take all the single particle states
- Make them antisymmetric

The filled lowest Landau level

Many-body wavefunction of the filled lowest Landau level

$$\Psi_{\mathrm{LLL}}(z_1, \cdots, z_{N_\mathrm{p}}) \propto \prod_{i<j}^{N_\mathrm{p}} (z_i - z_j) \exp\left[-\sum_{l}^{N_\mathrm{p}} |z_l|^2/(4\ell^2) \right]$$

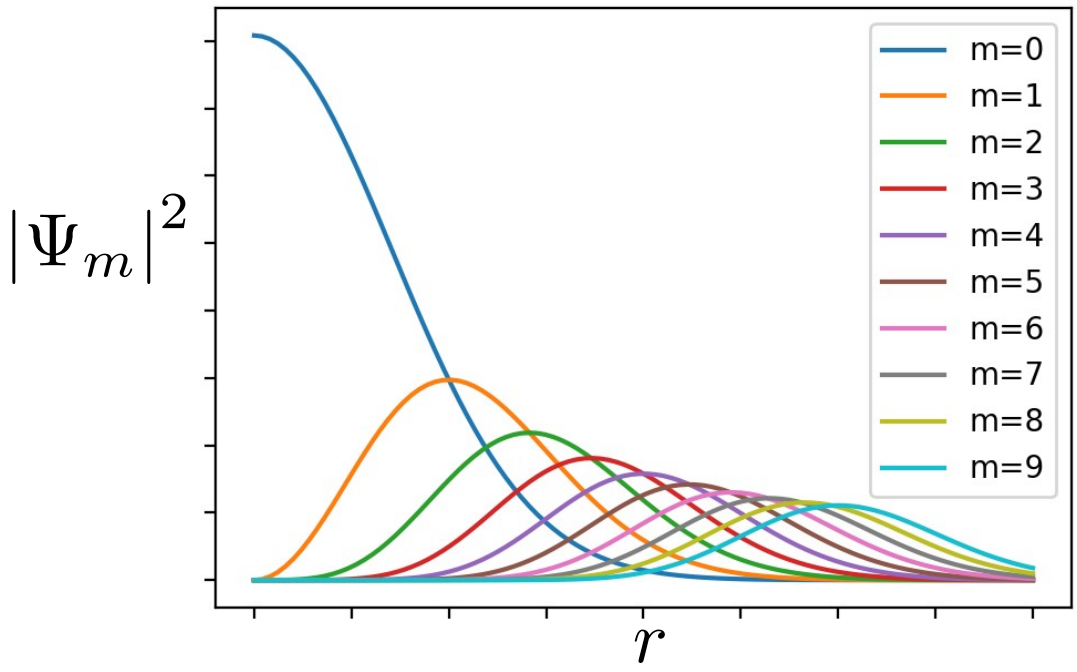
Fully antisymmetrical
(Pauli's principle)

Landau level like

$\ell \sim 1/\sqrt{B}$ $\qquad z_n = x_n + iy_n$

The Landau levels in the symmetric gauge

$$\Psi_m(z) \sim z^m e^{-|z|^2/(4\ell^2)}$$

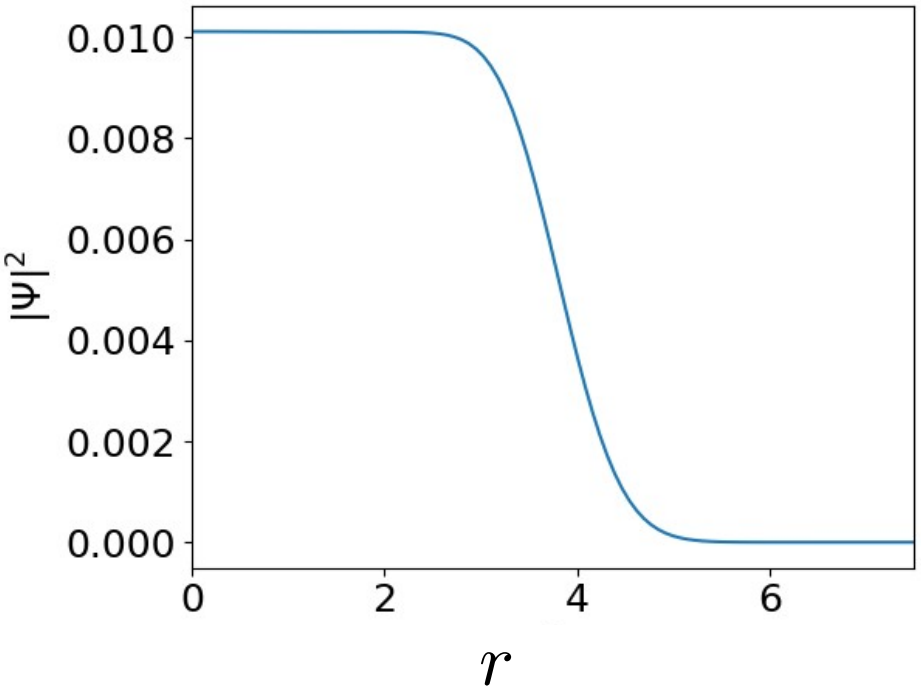


Maximum of the function at a radius

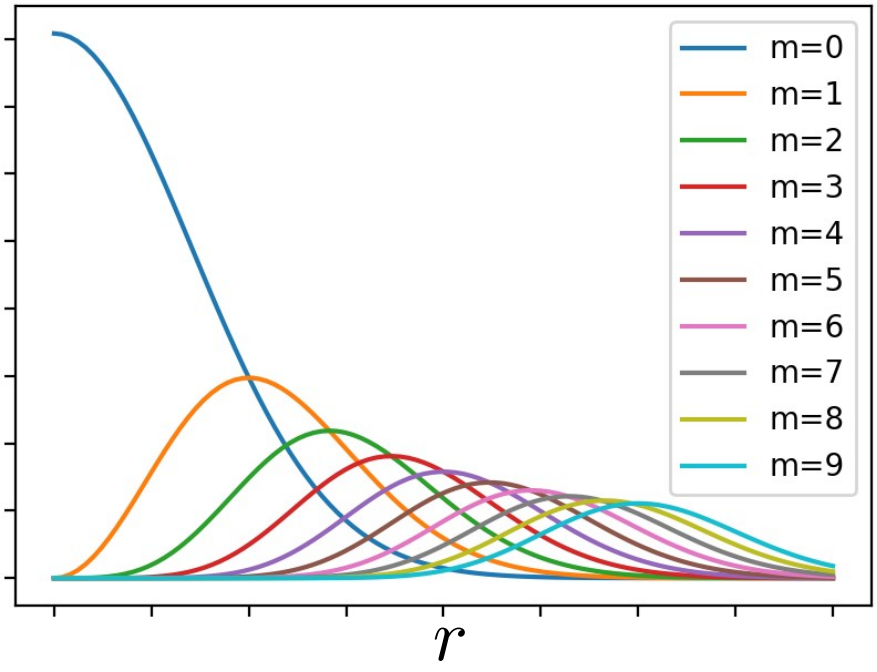
$$r = \ell\sqrt{2m}$$

The area enclosed by the Landau level is $\pi r^2 = 2\pi m\ell^2 \sim m/B$

The Landau levels in the symmetric gauge



$|\Psi_m|^2$



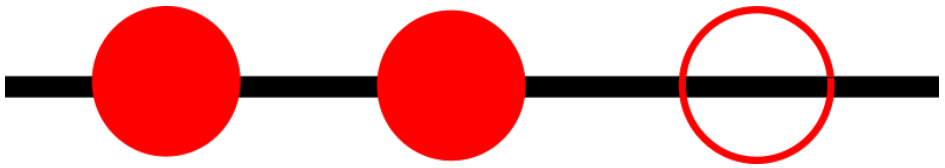
$$|\Psi|^2 = \sum_m |\Psi_m|^2$$

Constant total density, afterwards quickly going to zero

The wavefunction of a fractional quantum Hall state

Towards fractional filling

Let us take a fractional filling



Single particle wavefunction for the lowest Landau level

$$\Psi(z) \sim p(z) e^{-|z|^2/(4\ell^2)}$$

Polynomial

How do we write a wavefunction for fractional filling?

$$\Psi(z_1, z_2, \ldots, z_n)$$

Solving a Landau level with fractional filling

- What do we need
  - A many body wavefunction
  - That looks like a Landau level
  - That fulfills Pauli's exclusion principle
  - With the coordinates of the electrons highly correlated, avoiding each other (repulsive interactions)



The Laughlin's wave function

Let's generalize the Many-body LL wavefunction

$$\Psi^{(m)}_{\text{Laughlin}} = \prod_{i<j} (z_i - z_j)^m \prod_{i=1}^{N} e^{-|z_i|^2/\left(4\ell^2\right)}$$

$\ell \sim 1/\sqrt{B}$

Notice the m in the exponent

This wavefunction describes a Landau level with

$1/m$ electronic density

excitations with charge $1/m$

The Laughlin's wave function

$$\Psi^{(m)}_{\text{Laughlin}} = \prod_{i<j} (z_i - z_j)^m \prod_{i=1}^{N} e^{-|z_i|^2/\left(4\ell^2\right)}$$

Maximum power of the wavefunction

$$N_\phi = m(N-1)$$

Leading to a filling fraction

$$\nu = N/N_\phi \to 1/m$$

Quasiholes

From the many-body wavefunction, we can write down the form of a quasihole excitations with charge $1/n$

$$\Psi_{qh}(\mathbf{0}) = \left[\prod_{i=1}^{N} z_i\right] \Psi_{\text{Laughlin}}$$

(excitation at r=0)

$$\Psi_{qh}(w) = \left[\prod_{i=1}^{N} (z_i - w)\right] \Psi_{\text{Laughlin}}$$

(excitation at r=w)

Quasiholes

$$\Psi_{qh}(\mathbf{0}) = \left[\prod_{i=1}^{N} z_i\right] \Psi_{\text{Laughlin}}$$

$$m = 1/\nu$$

$$\Psi^{(m)}_{\text{Laughlin}} = \prod_{i<j} (z_i - z_j)^m \prod_{i=1}^{N} e^{-|z_i|^2/\left(4\ell^2\right)}$$

Increase the power of each filled orbital

Leaving an empty orbital with charge

$$e^* = \nu e$$

Quasiholes

Multiple quasiholes (M)

$$\Psi_{qhs}\left(w_1,\ldots,w_M\right)=\left[\prod_{\alpha=1}^{M}\prod_{i=1}^{N}\left(z_i-w_\alpha\right)\right]\Psi_{\mathrm{Laughlin}}$$

Some comments:
- Inserting n quasiholes is like putting a single hole
- It is a zero energy eigenstate (for the exact Hamiltonian with this wavefunction)
- **They have fractional braiding statistics**

Quasielectrons

$$\Psi_{qe}(\mathbf{0}) = \left(\prod_{i=1}^{N} \frac{\partial}{\partial z_i} \right] \phi \right) \prod_{i=1}^{N} e^{-|z_i|^2/\left(4\ell^2\right)}$$

$$m = 1/\nu$$

$$\Psi^{(m)}_{\text{Laughlin}} = \phi \prod_{i=1}^{N} e^{-|z_i|^2/\left(4\ell^2\right)}$$



Decrease the power of each filled orbital

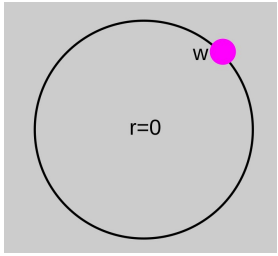
Filling an orbital with charge

$$e^* = \nu e$$

Fractional statistics in a nutshell

Recall the definition of a Berry phase

$$\Delta\gamma = -i \int_0^{2\pi} d\theta \, \langle \Psi(\theta) \, | \partial_\theta | \, \Psi(\theta) \rangle$$



Take a wavefunction with a single quasihole $\Psi(w) \sim \left[\prod_{i=1}^{N} (z_i - w) \right] \Psi_{\text{Laughlin}}^{(m)}$

$w = |w| e^{i\theta}$ And wind it around the origin, obtaining the phase

$$\Delta\gamma = 2\pi(1/m)\Phi/\phi_0 = \gamma_{AB}$$ *Aharonov-Bohm phase for charge e/m*

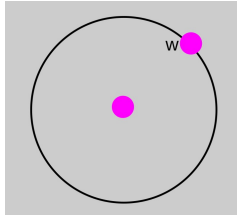
Enclosed flux

Quantum flux

Fractional statistics in a nutshell

Recall the definition of a Berry phase

$$\Delta\gamma = -i \int_0^{2\pi} d\theta \, \langle \Psi(\theta) \, | \partial_\theta | \, \Psi(\theta) \rangle$$



Take a wavefunction with two quasihole $\Psi(w) \sim \left[\prod_{i=1}^{N} (z_i)\right] \left[\prod_{i=1}^{N} (z_i - w)\right] \Psi^{(m)}_{\text{Laughlin}}$

$w = |w| e^{i\theta}$ And wind it around the origin, obtaining the phase

$$\Delta\gamma = \gamma_{AB} + \gamma_{\text{statistical}}$$

$\gamma_{\text{statistical}} = 2\pi/m$

Aharonov-Bohm phase

Statistical phase

The quantum Hall effect without Landau levels

Quantum Hall effect without Landau levels

- Net magnetic field is not necessary to have QH physics
- To have a non-zero Chern number, we just need to break time-reversal symmetry

Berry curvature

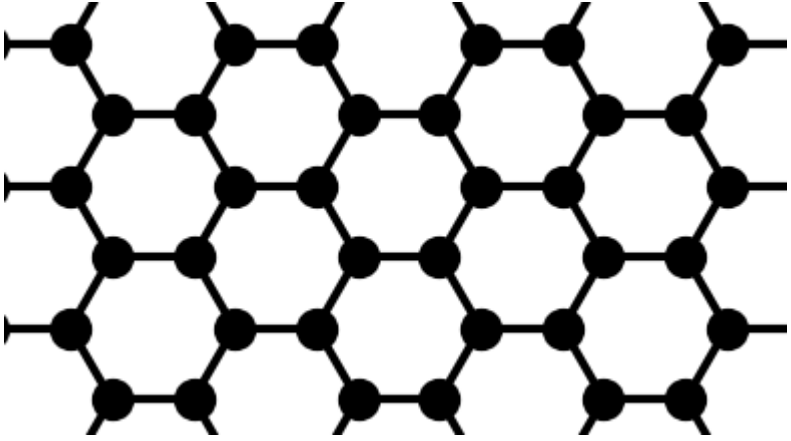
$$\Omega_\alpha = \partial_{k_x} A_y^\alpha - \partial_{k_y} A_x^\alpha$$

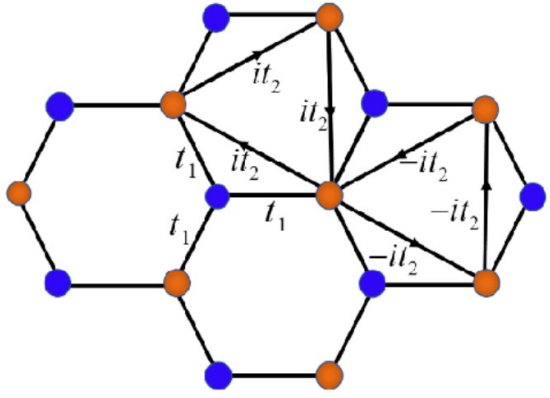
Chern number

$$\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k} = \sum_\alpha C_\alpha = C$$

The Haldane model

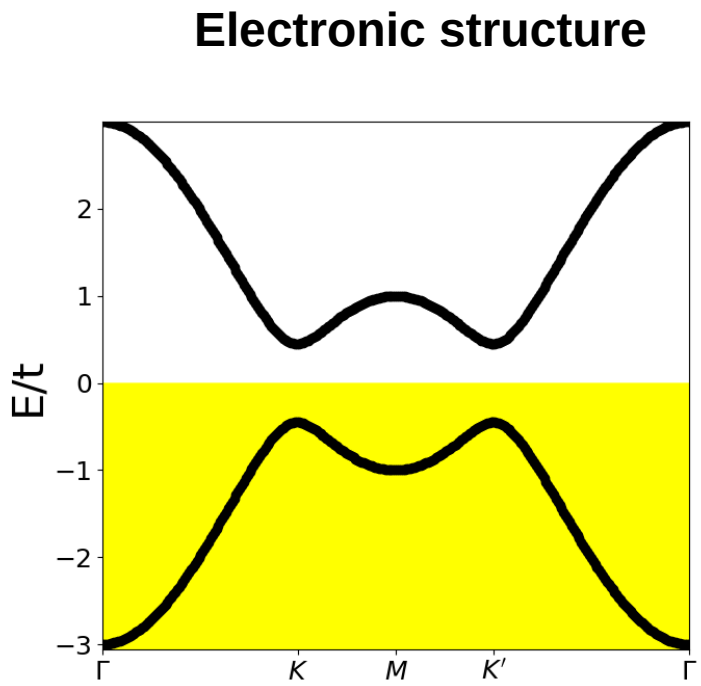
Take a model in the honeycomb lattice and include second-neighbor hopping breaking time-reversal symmetry

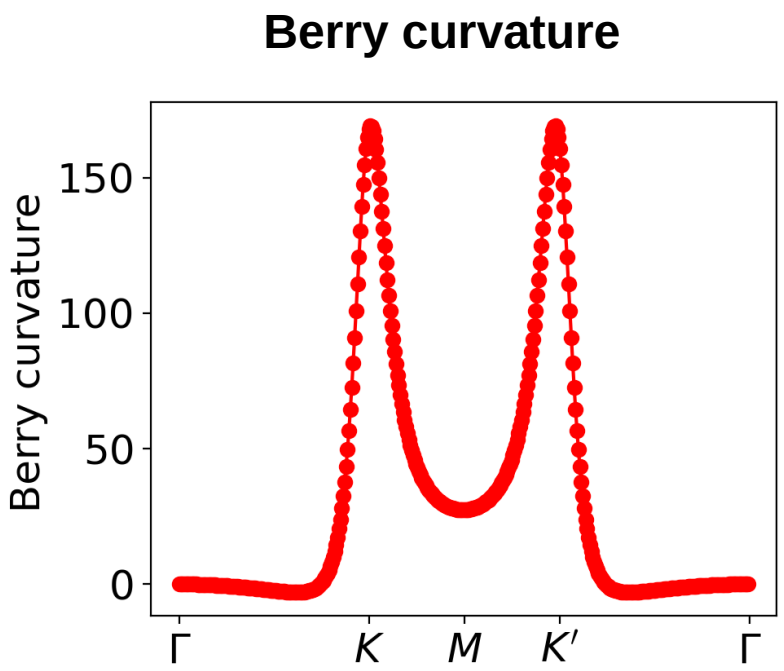




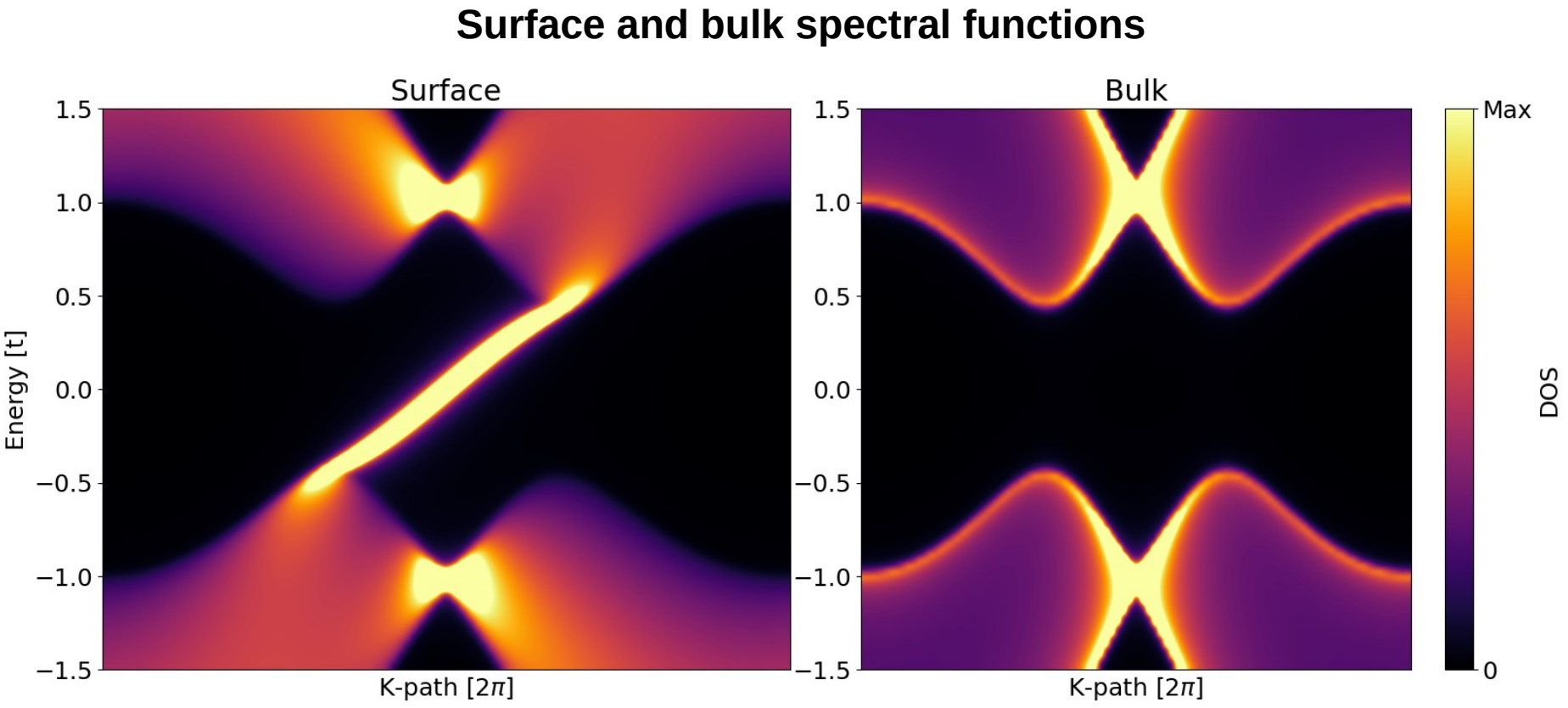
Equivalent to having a spatially dependent field that averages to zero in space

The Haldane model: bulk electronic structure

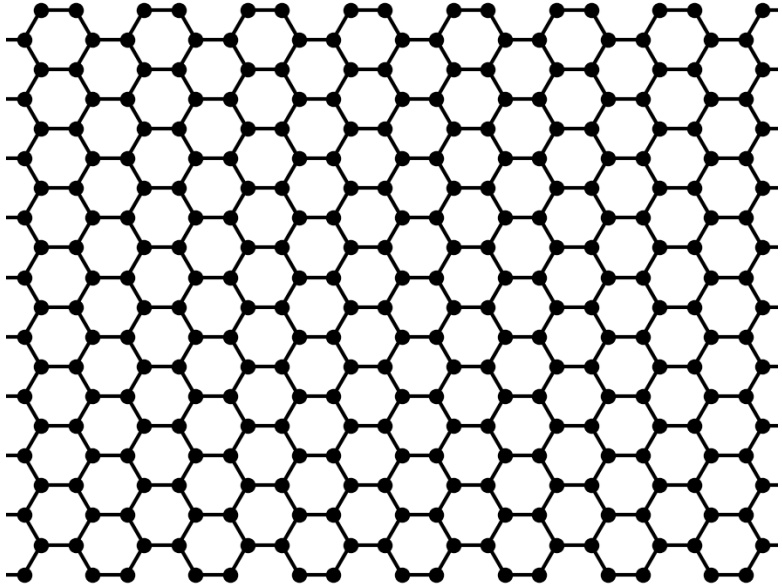


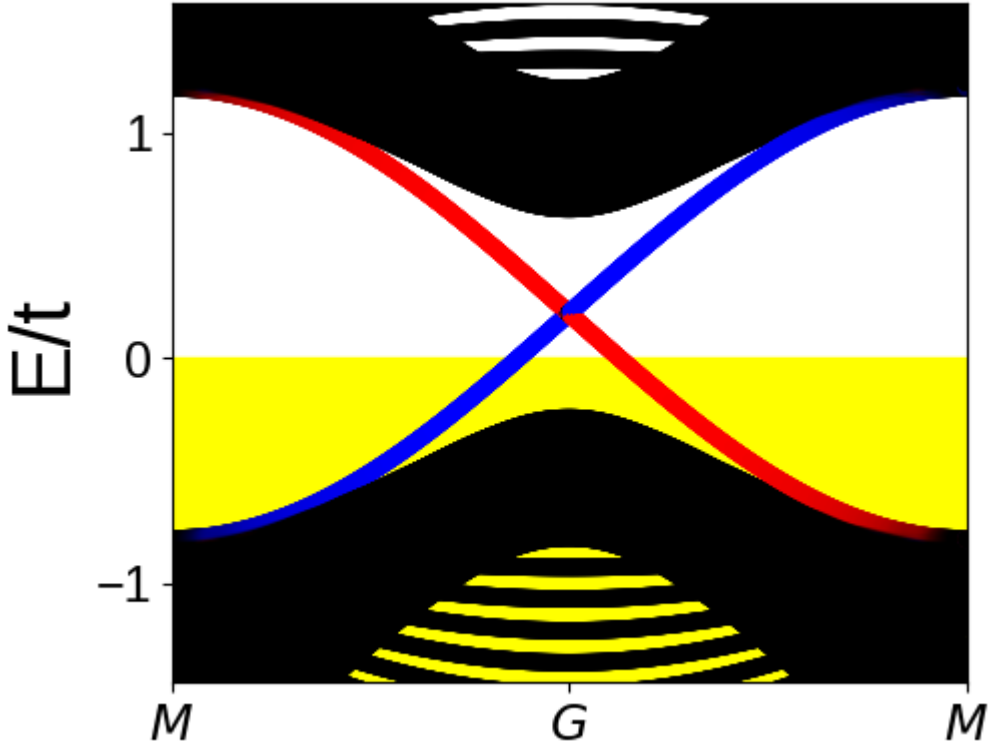


The Haldane model: edge electronic structure



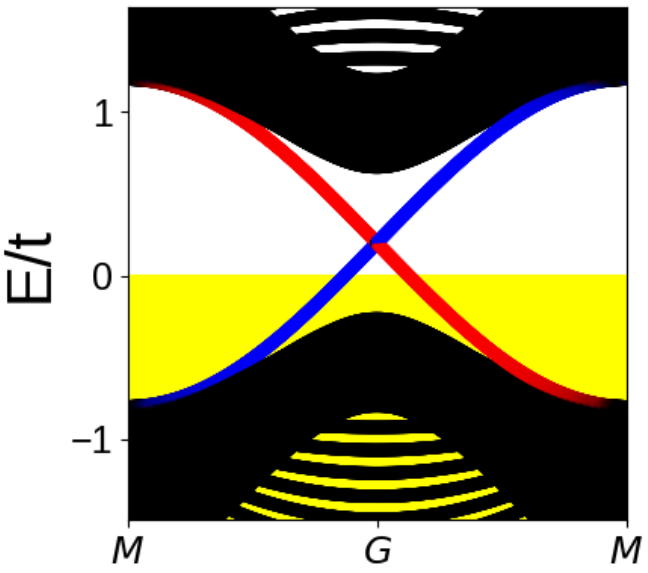
The Haldane model: ribbon electronic structure



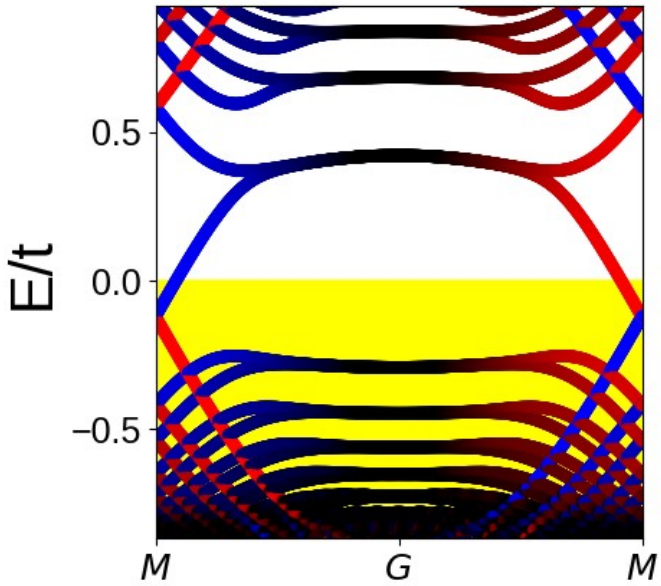


Topological equivalence between Haldane model and quantum Hall

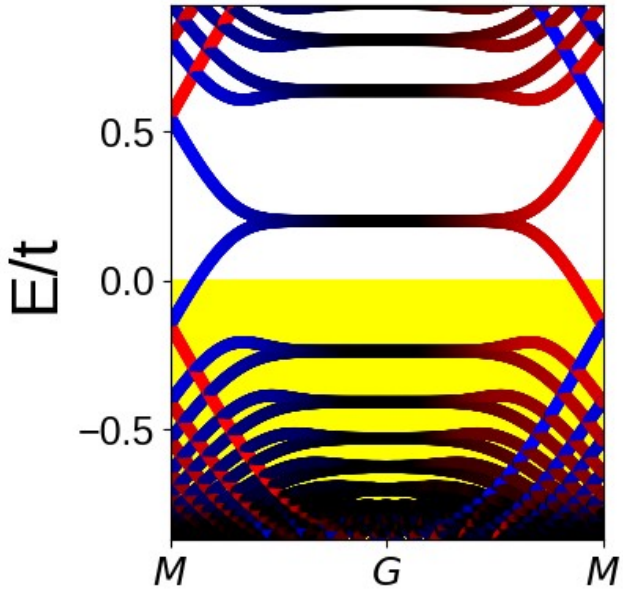
Haldane model



Haldane and quantum Hall



Quantum Hall

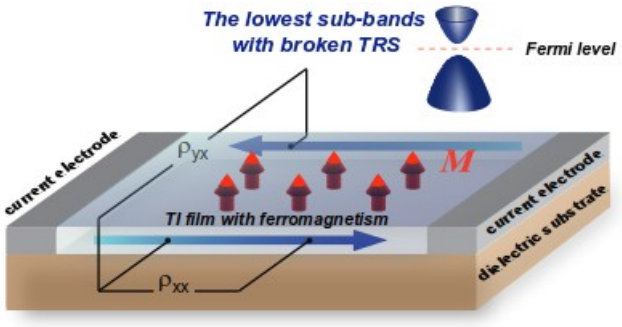


The bulk gap remains open as Haldane model is transformed in the quantum Hall state

Both Haldane model and quantum Hall have the same topological invariant

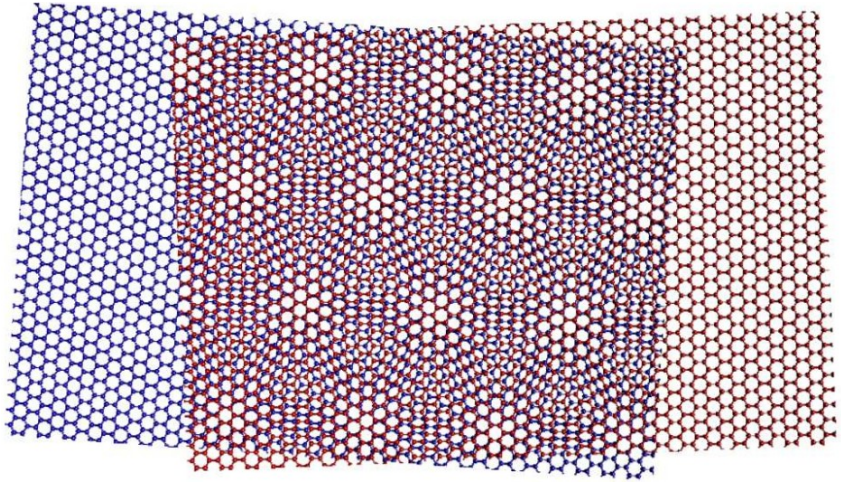
Real Chern insulating materials

Magnetically doped topological insulators



Cr-doped $(Bi,Sb)_2Te_3$

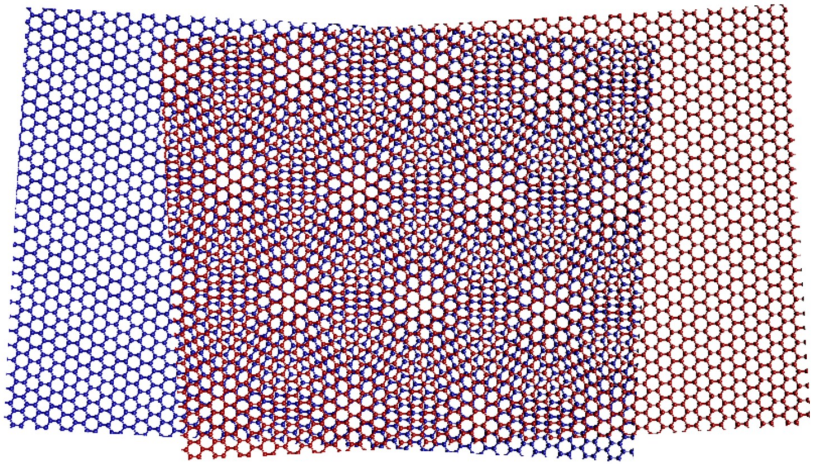
Twisted graphene bilayers



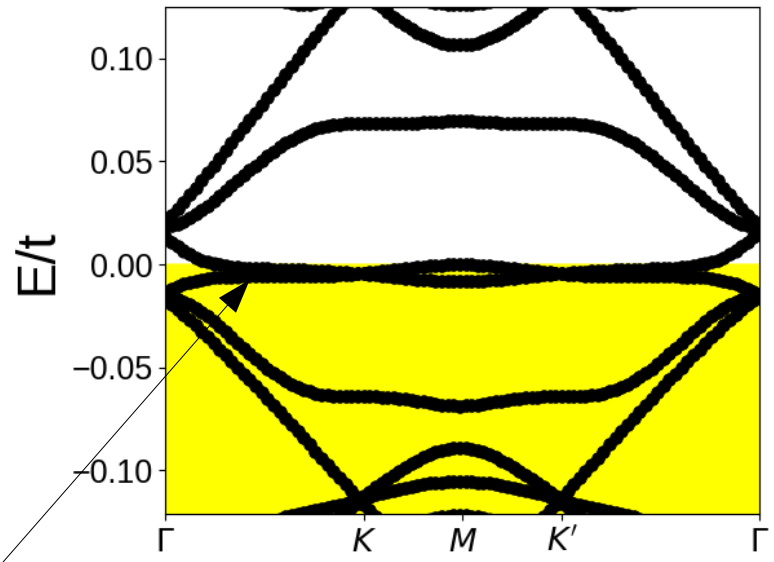
(aligned with BN substrate)

Real fractional Chern insulating materials

Twisted 2D materials



```
from pyqula import specialhamiltonian # special Hamiltonians library
h = specialhamiltonian.twisted_bilayer_graphene(13,ti=0.4) # TBG Hamiltonian
(k,e) = h.get_bands(num_bands=20) # compute band structure
```



(pseudo) Landau levels arising from an elastic gauge field

Twisted dichalcogenide materials ($MoTe_2$) show FQHE at zero magnetic field

Take home

- Quantum Hall effect can exist with and without external magnetic field
- Landau levels are topological flat bands of the quantum Hall state
- Correlations in a topological flat band give rise to fractional excitations